

Тренировка 15.04.2026

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$$Z = 11x_1 + 9x_2 \rightarrow \max$$

$$\begin{cases} 3x_1 + 4x_2 \leq 10 \\ 2x_1 + x_2 \leq 8 \\ x_i \geq 0, i=1, 2 \end{cases} \Rightarrow$$

$$3x_1 + 4x_2 + x_3 = 10$$

$$2x_1 + x_2 + x_4 = 8$$

$$x_i \geq 0, i=1, 2, 3, 4$$

$$\left(\begin{array}{cccc|c} 3 & 4 & 1 & 0 & 10 \\ 2 & 1 & 0 & 1 & 8 \end{array} \right)$$

Шаг 1:

b	1	x_1	x_2	x_3	x_4	CO
x_1	10	1	4	1	0	10/5
x_2	8	2	1	0	1	4
P	0	-11	-9	0	0	1000

$$X^1 = (0; 0; 10; 8)$$

Шаг 2:

b	1	x_1	x_2	x_3	x_4	CO
x_1	10/3	1	4/3	1/3	0	2
x_4	4/3	0	-5/3	-2/3	1	4/3
P	110/3	0	17/3	11/3	0	1000

$$X^2 = \left(\frac{10}{3}; 0; 0; \frac{4}{3} \right)$$

Ограничение не выполнено

$$\frac{1}{3}x_2 + \frac{1}{3}x_3 \geq \frac{1}{3} \Rightarrow \frac{1}{3}x_2 + \frac{1}{3}x_3 - x_5 = \frac{1}{3}$$

$$-\frac{1}{3}x_2 - \frac{1}{3}x_3 + x_5 = -\frac{1}{3}$$

Шаг 3:

b	1	x_1	x_2	x_3	x_4	x_5
x_1	10/3	1	4/3	1/3	0	0
x_4	4/3	0	-5/3	-2/3	1	0
x_5	-1/3	0	-1/3	1/3	0	1
P	110/3	0	17/3	11/3	0	0
CO	1000	1000	17	11	1000	1000

$$X^3 = \left(\frac{10}{3}; 0; 0; \frac{4}{3}; -\frac{1}{3} \right)$$

Problem 4:

b	1	x_1	x_2	x_3	x_4	x_5
x_1	3	1	2	0	0	2
x_2	2	0	1	0	1	5
x_3	1	0	1	1	0	-3
Z	33	0	2	0	0	2
CO	10	11	5	10	11	

$$X^4 = (3; 0; 1; 2; 0)$$

$$\frac{10}{3} - \frac{1}{3} \cdot \frac{-1}{3} = \frac{10}{3} - \frac{1}{3} = \frac{9}{3} = 3$$

$$\frac{4}{3} - \frac{-2}{3} \cdot \frac{-1}{3} = \frac{4}{3} + \frac{2}{3} = \frac{6}{3} = 2$$

$$\frac{110}{3} - \frac{-1}{3} \cdot \frac{11}{3} = \frac{110}{3} - \frac{11}{3} = \frac{99}{3} = 33$$

$$Z_{\max} = Z(3; 0) = 11 \cdot 3 + 9 \cdot 0 = 33$$

Answer: $Z_{\max} = Z(3; 0) = 33$

1.2

$$Z = 9x_1 + 13x_2 \rightarrow \max$$

$$\begin{cases} x_1 + 2x_2 \leq 10 \\ 3x_1 + 8x_2 \leq 33 \\ x_1 \geq 0, i = 1, 2 \end{cases}$$

$\Rightarrow$

$$\begin{cases} x_1 + 2x_2 + x_3 = 10 \\ 3x_1 + 8x_2 + x_4 = 33 \\ x_i \geq 0, i = 1, 2, 3, 4 \end{cases}$$

Table 1

b	1	x_1	x_2	x_3	x_4	CO
x_3	10	1	2	1	0	0
x_4	33	3	8	0	1	5
Z	0	-9	-23	0	0	$\frac{33}{8}$

4,125

$$X^1 = (10; 0; 10; 33)$$

Table 2

b	1	x_1	x_2	x_3	x_4	CO
x_3	$\frac{14}{8}$	$\frac{2}{8}$	0	1	$-\frac{1}{4}$	7
x_2	$\frac{33}{8}$	$\frac{3}{8}$	1	0	$\frac{1}{8}$	11
Z	$\frac{759}{8}$	$\frac{3}{8}$	0	0	$\frac{23}{8}$	

$$X^2 = \left(\frac{14}{8}; \frac{33}{8}; 0; 0 \right)$$

$$10 - \frac{2 \cdot 33}{8} = 10 - \frac{66}{8} = \frac{80 - 66}{8} = \frac{14}{8}$$

$$1 - \frac{2 \cdot 3}{8} = 1 - \frac{6}{8} = \frac{2}{8}$$

$$0 - \frac{2 \cdot 1}{8} = -\frac{1}{4}$$

$$0 - \frac{1 \cdot (-23)}{8} = \frac{23}{8}$$

$$-9 - \frac{3 \cdot (-23)}{8} = -9 + \frac{69}{8} = \frac{-72 + 69}{8} = -\frac{3}{8}$$

$$0 - \frac{33 \cdot (-23)}{8} = \frac{759}{8}$$

Matr. 3

b	1	x_1	x_2	x_3	x_4	CO
x_1	7	1	0	1	-1	7
x_2	$\frac{3}{2}$	0	1	$-\frac{3}{2}$	$\frac{1}{2}$	$\frac{5}{2}$
Σ	$\frac{195}{2}$	0	0	$\frac{3}{2}$	$\frac{5}{2}$	

$$X^3 = (7; \frac{3}{2}; 0; 0)$$

Prüfungsausschuss 192

$$\frac{33}{8} - \frac{19}{8} \cdot \frac{3}{8} = \frac{33}{8} - \frac{57}{8} = \frac{33 - 57}{8} = \frac{-24}{8} = -3$$

$$\frac{759}{8} - \frac{19}{8} \cdot \frac{3}{8} = \frac{759}{8} - \frac{57}{8} = \frac{759 - 57}{8} = \frac{702}{8} = \frac{351}{4}$$

$$0 - 1 \cdot \frac{3}{8} = -\frac{3}{8}$$

$$0 - 1 \cdot \frac{3}{8} = -\frac{3}{8}$$

$$\frac{23}{8} - \frac{1}{8} \cdot \frac{3}{8} = \frac{23}{8} - \frac{3}{8} = \frac{20}{8} = \frac{5}{2}$$

Task 4:

b	1	x_1	x_2	x_3	x_4	x_5
x_1	7	1	0	4	-1	0
x_2	$\frac{3}{2}$	0	1	$\frac{3}{2}$	$\frac{1}{2}$	0
x_3	$-\frac{1}{2}$	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	1
Z	$\frac{195}{2}$	0	0	$\frac{3}{2}$	$\frac{5}{2}$	0
CO	111	111	111	3	5	111

$$\frac{1}{2} x_3 + \frac{1}{2} x_4 \geq \frac{1}{2}$$

$$-\frac{1}{2} x_3 - \frac{1}{2} x_4 - x_5 = -\frac{1}{2}$$

$$X^4 = (7; \frac{3}{2}; 0; 0; -\frac{1}{2})$$

Task 5:

b	1	x_1	x_2	x_3	x_4	x_5
x_1	3	1	0	0	5	2
x_2	3	0	1	0	5	5
x_3	1	0	0	1	1	-2
Z	96	0	0	0	2	2
CO	111	111	111	111	2	2

$$7 - 4 \cdot -\frac{1}{2} = 7 - 4 = 3$$

$$\frac{3}{2} - \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{2} + \frac{1}{4} = \frac{6}{4} + \frac{1}{4} = \frac{7}{4}$$

$$\frac{195}{2} - \frac{1}{2} \cdot \frac{3}{2} = \frac{195}{2} - \frac{3}{4} = \frac{192}{2} = 96$$

$$X^6 = (3; 3; 1; 0; 0)$$

$$Z_{max} = Z(3, 3) = 9 \cdot 3 + 25 \cdot 3 = 27 + 75 = 102$$

Given: $Z_{\max} = Z(3, 3) = 96$

$\sqrt{3}$

$Z = x_1 + x_2 \rightarrow \max$

$$\begin{cases} 2x_1 + x_2 \leq 5 \\ 2x_1 + 3x_2 \leq 9 \\ x_i \geq 0, i=1, 2 \end{cases} \Rightarrow \begin{cases} 2x_1 + x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + x_4 = 9 \\ x_i \geq 0, i=1, 2, 3, 4 \end{cases}$$

$$\left(\begin{array}{cccc|c} 2 & 1 & 1 & 0 & 5 \\ 2 & 3 & 0 & 1 & 9 \end{array} \right)$$

Table 1

b.r.	1	x_1	x_2	x_3	x_4	CO
x_3	5	2	1	1	0	$\frac{5}{2}$
x_4	9	2	3	0	1	$\frac{9}{2}$
Z	0	-1	-1	0	0	

Table 2:

b.r.	1	x_1	x_2	x_3	x_4	CO
x_1	$\frac{5}{2}$	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	5
x_4	4	0	2	-1	1	2
Z	$\frac{5}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	

Table 3

b.r.	1	x_1	x_2	x_3	x_4	CO
x_1	$\frac{3}{2}$	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	2
x_2	2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	2
Z	$\frac{7}{2}$	0	0	$\frac{1}{4}$	$\frac{1}{4}$	

Iteration 4:

δ, R	1	x_1	x_2	x_3	x_4	x_5
x_1	$\frac{5}{2}$	1	0	$\frac{3}{2}$	$-\frac{1}{4}$	0
x_2	2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	0
x_3	$-\frac{1}{2}$	0	0	$-\frac{3}{4}$	$-\frac{3}{4}$	1
Z	$\frac{7}{2}$	0	0	$\frac{1}{4}$	$\frac{1}{4}$	0
CO	10	10	10	$\frac{1}{3}$	$\frac{2}{3}$	10

Iteration 5:

δ, R	1	x_1	x_2	x_3	x_4	x_5
x_1	1	1	0	0	-1	1
x_2	$\frac{7}{3}$	0	1	0	1	$-\frac{2}{3}$
x_3	$\frac{2}{3}$	0	0	1	1	$-\frac{1}{3}$
Z	$\frac{10}{3}$	0	0	0	0	$\frac{1}{3}$
CO	10	10	10	10	10	10

$$\left(\frac{1}{3} < \frac{2}{3} \right)$$

Iteration 6:

δ, R	1	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1	1	0	0	-1	1	0
x_2	$\frac{7}{3}$	0	1	0	1	$-\frac{2}{3}$	0
x_3	$\frac{2}{3}$	0	0	1	1	$-\frac{4}{3}$	0
x_4	$-\frac{2}{5}$	0	0	0	0	$-\frac{2}{3}$	1
Z	$\frac{10}{3}$	0	0	0	0	$\frac{1}{3}$	0
CO	10	10	10	10	-	$\frac{1}{3}$	10

Matr 7:

δ_i	1	x_1	x_2	x_3	x_4	x_5	x_6
x_1	0	1	0	0	-1	0	$\frac{3}{2}$
x_2	3	0	1	0	$\frac{1}{2}$	0	-1
x_3	2	0	0	1	$\frac{1}{2}$	0	-2
x_5	1	0	0	0	0	1	$-\frac{5}{2}$
Z	3	0	0	0	0	0	$\frac{1}{2}$
CO	1	1	1	1	-	1	-

$$X^7 = (0; 3; 2; 0; 1)$$

Matr 8

δ_i	1	x_1	x_2	x_3	x_4	x_5	x_6
x_1	2	2	2	2	2	2	2
x_2	1	2	2	2	2	2	2
x_4	2	2	2	2	2	2	2
x_5	1	2	2	2	2	2	2
Z	3	0	0	0	0	0	$\frac{1}{2}$

$$Z_{\max} = Z(0, 3) = 0 + 3 = 3$$

$$Z_{\max} = Z(1, 2) = 1 + 2 = 3$$

$$Z_{\max} = Z(2, 1) = 2 + 1 = 3$$

Umkehr: $Z_{\max} = Z(0, 3) = Z(1, 2) = Z(2, 1) = 3$

Целевая функция $\sqrt{7}$
 $\sqrt{4}$

$$Z = 24x_1 + 16x_2 \rightarrow \max$$

$$\begin{cases} 2x_1 + x_2 \leq 20 \\ 5x_1 + 4x_2 \leq 50 \\ x_i \geq 0, i = 1, 2 \end{cases} \Rightarrow \begin{cases} 2x_1 + x_2 + x_3 = 20 \\ 5x_1 + 4x_2 + x_4 = 50 \\ x_i \geq 0, i = 1, 2, 3, 4 \end{cases}$$

Шаг 1:

б.п.	1	x_1	x_2	x_3	x_4	св
x_3	20	(2)	1	1	0	10
x_4	50	5	4	0	1	10,4
Z	0	-24	-16	0	0	0

Шаг 2:

б.п.	1	x_1	x_2	x_3	x_4	св
x_1	10	1	$\frac{1}{2}$	$\frac{1}{2}$	0	20
x_2	2	0	($\frac{3}{2}$)	$-\frac{5}{2}$	1	$\frac{4}{3}$
Z	240	0	$-\frac{11}{2}$	$\frac{21}{2}$	0	0

Шаг 3:

б.п.	1	x_1	x_2	x_3	x_4	св
x_1	$\frac{28}{3}$	1	0	$\frac{4}{3}$	$-\frac{1}{3}$	10
x_2	$\frac{4}{3}$	0	1	$-\frac{5}{3}$	$\frac{2}{3}$	$\frac{10}{3}$
Z	$\frac{652}{3}$	0	0	$\frac{4}{3}$	$\frac{11}{3}$	0

Table 4:

		x_1	x_2	x_3	x_4	x_5
$\bar{b}$	1	x_1	x_2	x_3	x_4	x_5
x_1	$\frac{28}{3}$	1	0	$\frac{4}{3}$	$-\frac{1}{3}$	0
x_2	$\frac{4}{3}$	0	1	$-\frac{5}{3}$	$\frac{2}{3}$	0
x_3	$-\frac{1}{3}$	0	0	$-\frac{1}{3}$	$-\frac{2}{3}$	1
x_4	$\frac{11}{3}$	0	0	$\frac{4}{3}$	$\frac{11}{3}$	0
Z	$\frac{152}{3}$	0	0	$\frac{4}{3}$	$\frac{11}{3}$	0
CO	211	211	211	4	11	211

Table 5:

		x_1	x_2	x_3	x_4	x_5
$\bar{b}$	1	x_1	x_2	x_3	x_4	x_5
x_1	8	1	0	0	2	2
x_2	3	0	1	0	4	4
x_3	2	0	0	1	2	4
Z	216	0	0	0	1	4
CO	211	211	211	211	211	211

$$Z_{\max} = Z(8, 3) = 21 \cdot 8 + 16 \cdot 3 = 168 + 48 = 216$$

Optimal: $Z_{\max} = Z(8, 3) = 216$

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$$Z = 16x_1 + 41x_2 \rightarrow \max$$

$$\begin{cases} x_1 + 2x_2 \leq 29 \\ 3x_1 + 8x_2 \leq 85 \\ x_i \geq 0, i=1,2 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 29 \\ 3x_1 + 8x_2 + x_4 = 85 \\ x_i \geq 0, i=1,2,3,4 \end{cases}$$

Table 1:

f.r.		x_1	x_2	x_3	x_4	CO
x_5	24	1	2	1	0	10
x_2	85	3	8	0	1	10 $\frac{5}{8}$
Z	0	-16	-44	0	0	224

Table 2:

f.r.		x_1	x_2	x_3	x_4	CO
x_2	$\frac{1}{8}$	$\frac{2}{8}$	1	$-\frac{2}{8}$	11	
x_1	$\frac{85}{8}$	$\frac{3}{8}$	1	$\frac{1}{8}$	28	
Z	$\frac{3485}{8}$	$-\frac{5}{8}$	0	$\frac{41}{8}$	224	

Table 3:

f.r.		x_1	x_2	x_3	x_4	CO
x_1	11	1	0	4	-1	2
x_2	$\frac{13}{2}$	0	1	$-\frac{3}{2}$	$\frac{1}{2}$	4
Z	$\frac{885}{2}$	0	0	$\frac{5}{2}$	$\frac{9}{2}$	224

Table 4:

f.r.		x_1	x_2	x_3	x_4	x_5
x_1	11	1	0	4	-1	0
x_2	$\frac{13}{2}$	0	1	$-\frac{3}{2}$	$\frac{1}{2}$	0
x_5	$\frac{1}{2}$	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	1
Z	$\frac{885}{2}$	0	0	$\frac{5}{2}$	$\frac{9}{2}$	0
CO	224	224	224	5	9	224

Task 5:

		x_1	x_2	x_3	x_4	x_5
r.h.s.	1					
x_1	7	1	0	0	2	2
x_2	8	0	1	0	2	2
x_3	1	0	0	1	1	-2
Z	440	0	0	0	2	5
CV	16	16	44		2	5

$$Z_{\max} = Z(7, 8) = 16 \cdot 7 + 44 \cdot 8 = 112 + 352 = 464$$

Answer: $Z_{\max} = Z(7, 8) = 464$